

Para

Wireless data communication and mobility involve several elements across different networks like WiFi, 4G, and 5G. Key components include:

- **Wireless hosts:** End devices (smartphones, tablets, laptops, IoT devices) that run applications. They may or may not be mobile.
- **Wireless links:** Connections between hosts and base stations (or other hosts) via wireless communication links. These links vary in **transmission rate** and **coverage range**, influenced by distance, channel conditions, and number of users.

Later sections of the chapter will discuss specific architectures, error characteristics, and standards in detail.

- **Wireless links:** Connect wireless hosts (like laptops, phones, IoT devices) to the larger network infrastructure. They can also connect routers, switches, and other network equipment.
- **Base station:**
 - A central part of wireless network infrastructure.
 - Unlike wired networks, it has no direct counterpart.
 - Responsible for sending/receiving data (packets) between hosts and the larger network.
 - Coordinates communication of multiple wireless hosts.
 - A host is *associated* with a base station if:
 1. It is within the base station's communication range.
 2. It uses the base station to relay data to/from the larger network.
 - Examples: **Cell towers** (cellular networks) and **Access points** (WiFi).

Para

- **Base station role:** Connects wireless hosts to the larger network (Internet, corporate, home) as a link-layer relay.
- **Infrastructure mode:** Hosts rely on the base station for services (routing, address assignment, DNS, etc.).
- **Ad hoc mode:** No infrastructure; hosts themselves provide routing, addressing, name translation, etc.
- **Mobility & handoff:** When a host moves out of one base station's range into another, it switches connection (handoff/handover). Raises challenges like location tracking, addressing, and maintaining ongoing connections (e.g., TCP, calls).
- **Types of wireless networks:**
 1. **Single-hop, infrastructure-based** – Base station connects directly to wired network; communication is host ↔ base station (e.g., WiFi in cafés, 4G LTE).
 2. **Single-hop, infrastructure-less** – No base station; nodes communicate directly, possibly coordinated by one node (e.g., Bluetooth).
 3. **Multi-hop, infrastructure-based** – Base station exists, but some hosts relay through others to reach it (e.g., sensor networks, mesh networks).
 4. **Multi-hop, infrastructure-less** – No base station; nodes relay among themselves, often mobile (e.g., MANETs, VANETs).
- **Focus:** This chapter mainly covers single-hop, infrastructure-based networks.

Wireless Links and Network Characteristics

- **Decreasing signal strength (path loss):** Signals weaken over distance and when passing through objects.
- **Interference:** Signals from other devices in the same frequency band (e.g., 2.4 GHz phones, microwaves) disrupt transmission; newer standards use 5 GHz to reduce this.
- **Multipath propagation:** Signals reflect off objects and arrive at different times, causing distortion; movement of objects can change this effect over time.

SNR (Signal to Noise Ratio)



What SNR Means

- **Definition:** SNR is a **relative measure** comparing the strength of the desired signal (the actual data you want) to the strength of background noise (unwanted interference).
- **Units:** Measured in **decibels (dB)** — calculated as
$$20 \times \log_{10}(\text{signal amplitude} \div \text{noise amplitude}).$$
- **Intuition:** The higher the SNR, the easier it is for the receiver to correctly extract the transmitted data from the “static.”



Why It Matters in Wireless Links

Wireless channels suffer from:

- **Attenuation (path loss)** – signal fading over distance or through materials.

- **Interference** – from devices on the same frequency (e.g., microwaves, phones).
- **Multipath propagation** – signal reflections causing distortion.

SNR captures the *net effect* of all that degradation **plus** environmental noise.

A *low* SNR means the useful signal is buried in noise, increasing the likelihood of **bit errors**.



SNR, BER & Modulation — The Trade-Off

From Figure 7.3 in your text:

Condition	Effect on Bit Error Rate (BER)
Higher SNR	Lower BER — fewer bits get corrupted.
Lower SNR	Higher BER — errors become frequent.
Faster modulation at same SNR	Higher BER (speed costs reliability).
Adaptive modulation	Dynamically chooses fastest scheme that still meets BER constraints.

Example:

- At **10 dB SNR**:
 - *BPSK* at 1 Mbps → $\text{BER} < 10^{-7}$ (excellent reliability).
 - *QAM16* at 4 Mbps → $\text{BER} \approx 10^{-1}$ (unusable).
 - Increase SNR to **20 dB** → *QAM16* now achieves $\text{BER} \approx 10^{-7}$, making it a good high-speed choice.
-

Practical Engineering Balance

- Increasing **transmit power** → boosts SNR, reduces BER... but drains batteries and risks interfering with others.
 - Adaptive systems (like Wi-Fi, 4G, 5G) constantly monitor SNR to select the **best modulation & coding** for the current channel.
-

If you want, I can map this to **Python code** that, given a target BER, picks an appropriate modulation scheme based on SNR thresholds — which could be a neat link between the theory here and your programming strengths.

Shall I prepare that?

What Are Bit Errors?

- A **bit error** happens when a bit sent by a transmitter is **received incorrectly** at the receiver — for example, a 1 flips to 0 or vice-versa. It's essentially a single-bit corruption in the data stream.

Bit Errors, SNR, and BER in Wireless Links

- **Bit errors in wireless links:** More common than in wired links due to impairments like attenuation, interference, and multipath propagation.
- **Protocols to handle errors:** Wireless protocols (e.g., 802.11) use:
 - **CRC error detection codes** (to detect corrupted frames).
 - **Link-level reliable data transfer** (retransmits corrupted frames).
- **Received signal at host:** A combination of:

- Degraded original signal (weakened by attenuation, multipath, etc.).
- Background environmental noise.
- **Signal-to-Noise Ratio (SNR):**
 - A measure of received signal strength relative to noise.
 - Higher SNR = easier to extract transmitted signal.
 - Measured in **decibels (dB)** = $20 \times \log_{10}(\text{signal amplitude} \div \text{noise amplitude})$.
- **Relation to Bit Error Rate (BER):**
 - BER = probability that a transmitted bit is received incorrectly.
 - BER decreases as SNR increases.
 - BER also depends on the **modulation technique** used for encoding.
- **Figure 7.3:** Shows BER vs SNR for three modulation techniques on an idealized wireless channel.
- **Note:** Detailed theory of modulation, coding, and BER analysis is beyond the book's scope.

Physical-Layer Characteristics in Wireless Communication

- **SNR vs BER and Transmission Power**
 - For a given modulation scheme: Higher **Signal-to-Noise Ratio (SNR)** → Lower **Bit Error Rate (BER)**.

- Sender can increase SNR by raising transmission power → reduces probability of frame errors.
- Limitations of increasing power:
 - Minimal gain beyond a certain BER threshold (e.g., reducing BER from 10^{-12} to 10^{-13} has little practical impact).
 - Higher energy consumption (critical for battery-powered devices).
 - Increased interference with other senders' transmissions.
- **SNR, Modulation Technique, and Transmission Rate**
 - For a fixed SNR: Higher bit-rate modulation = higher BER.
 - Example from Figure 7.3:
 - **At 10 dB SNR:**
 - **BPSK (1 Mbps)** → BER < 10^{-7} (very reliable).
 - **QAM16 (4 Mbps)** → BER $\approx 10^{-1}$ (too high, not useful).
 - **At 20 dB SNR:**
 - **QAM16 (4 Mbps)** → BER $\approx 10^{-7}$ (acceptable, preferred due to higher rate).
 - **BPSK (1 Mbps)** → BER extremely low, but slower.
 - Conclusion: Choice of modulation depends on trade-off between transmission rate and acceptable BER.
- **Adaptive Modulation and Coding**
 - Modulation technique should adapt dynamically to channel conditions (mobility, environment changes).

- Used in **WiFi (802.11)** and **4G/5G cellular networks**.
- Goal: Select modulation that maximizes transmission rate while keeping BER within acceptable limits.
- **Hidden Terminal Problem**
 - Occurs when two senders (A and C) cannot detect each other's transmissions due to obstacles (e.g., walls, mountains), yet both interfere at a common receiver (B).
 - Results in collisions at B, even though A and C are "hidden" from each other.
- **Signal Fading and Undetectable Collisions**
 - Even without obstacles, signals weaken (fade) with distance.
 - Scenario: A and C are too far to sense each other's transmissions but still strong enough to interfere at receiver B.
 - Illustrated in Figure 7.4(b).
- **Key Difference vs Wired Networks**
 - In **wired broadcast links**: All nodes hear all transmissions (collisions detectable).
 - In **wireless links**: Hidden terminals and fading make **collision detection and multiple access control much more complex**.

1 Bit Rate — How Fast You're Sending Data

- **Definition:** The number of bits transmitted per second (bps). Example: 1 Mbps = 1 million bits per second.
- **What affects it:**
 - The **modulation scheme** (how many bits you pack into each transmitted symbol).

- The **symbol rate** (how many symbols you can send per second over the channel).

2 Modulation — How You Encode Bits on a Wave

- **Definition:** The process of mapping digital bits to changes in an analog carrier wave (e.g., altering its amplitude, frequency, or phase).
- **Common examples** from the textbook:
 - **BPSK** (Binary Phase-Shift Keying) — 1 bit per symbol.
 - **QAM16** (Quadrature Amplitude Modulation, 16-state) — 4 bits per symbol.
 - **QAM256** — 8 bits per symbol.

Hidden Terminal Problem

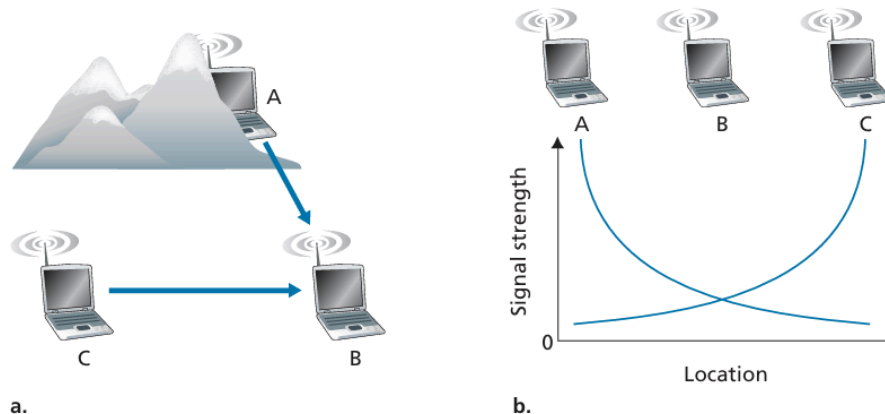


Figure 7.4 ♦ Hidden terminal problem caused by obstacle (a) and fading (b)

Suppose that Station A is transmitting to Station B. Suppose also that Station C is transmitting to Station B. With the so-called hidden terminal problem, physical obstructions in the environment (for example, a mountain or a building) may prevent A and C from hearing each other's transmissions, even though A's and C's transmissions are indeed interfering at the destination, B. This is shown in Figure 7.4(a). A second scenario that results in undetectable collisions at the

receiver results from the fading of a signal's strength as it propagates through the wireless medium. Figure 7.4(b) illustrates the case where A and C are placed such that their signals are not strong enough to detect each other's transmissions, yet their signals are strong enough to interfere with each other at station B. As we'll see in Section 7.3, the hidden terminal problem and fading make multiple access in a wireless network considerably more complex than in a wired network

7.2 CDMA(Code Division Multiple Access)

 Code Division Multiple Access (CDMA) – Youtube Link

And Book. No content here

Cases When Codes Shouldn't Work

1. Codes are identical (or negatives of each other)

- If two senders use the **same spreading code**, their signals are **indistinguishable** at the receiver.
- Receiver ends up getting the **sum of bits** instead of separating them.

Example:

- Sender 1 code: $c(1)=(1,1)c^{\{(1)\}}=(1,1)$
- Sender 2 code: $c(2)=(1,1)c^{\{(2)\}}=(1,1)$

Channel: $y=d_1(1,1)+d_2(1,1)=(d_1+d_2)(1,1)$ $y = d_1(1,1)+d_2(1,1)=(d_1+d_2)(1,1)$.

- If $d_1=+1, d_2=-1$ $d_1=+1, d_2=-1 \rightarrow y=(0,0)$.

- Receiver sees **zero**, so both bits are lost.

✗ **Does not work.**

2. Codes are not orthogonal (cross-correlation $\neq 0$)

- If codes are different but **not orthogonal**, one user's signal leaks into the other's detection.
- This creates **interference**, which may cause wrong bit decisions.

Example:

- Sender 1 code: $c(1) = (1, 1, 1, 1)$ $c^{\{1\}} = (1, 1, 1, 1)$
- Sender 2 code: $c(2) = (1, 1, 1, -1)$ $c^{\{2\}} = (1, 1, 1, -1)$

Dot product: $1+1+1-1=2 \neq 0$ $1+1+1-1=2 \neq 0$.

So, when receiver 1 correlates with $c(1)c^{\{1\}}$:

$$\hat{d}_1 = d_1 + 0.5 d_2$$

If $d_1 = -1, d_2 = +1$ $\hat{d}_1 = -1, \hat{d}_2 = +1$, then $\hat{d}_1 = -0.5$.

This might be misclassified (depending on noise).

✗ **Risk of decoding errors.**

3. Codes are too short (not enough spreading)

- If the spreading code length M is very small (like 2 chips), even if codes are "orthogonal," random noise or timing misalignment can cause large errors.
 - Practical CDMA systems use **longer codes** (e.g., 64, 128 chips) to reduce error probability.
-

4. Hidden terminals or multipath effects

- Even if codes are orthogonal **in theory**, in practice wireless effects (fading, multipath, imperfect synchronization) break orthogonality.
 - Two codes that should cancel out might not perfectly cancel, so interference leaks in.
✗ Real-world CDMA systems require strict synchronization.
-

✓ Summary

Codes **don't work** when:

1. They are **identical/negatives** → users become indistinguishable.
 2. They are **not orthogonal** → cross-correlation $\neq 0$ → interference.
 3. They are **too short** → weak spreading, poor robustness.
 4. **Practical wireless issues** (multipath, async clocks) ruin orthogonality.
-

👉 Do you want me to **build a small numeric example with 2 users using non-orthogonal codes** and show exactly how one user's bit corrupts the other during decoding?

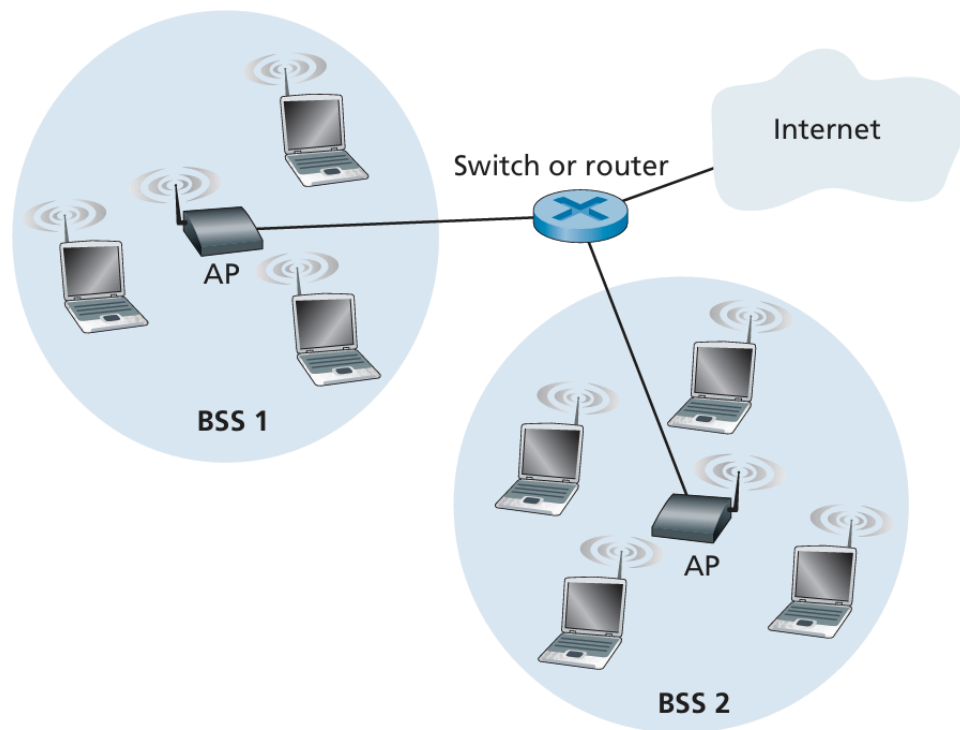
7.3 WiFi: 802.11 Wireless LANs

Summary of IEEE 802.11 (WiFi) Standards

- **Overview:** IEEE 802.11 (WiFi) is the dominant WLAN standard since the 1990s, used for home, office, and business networks.
- **Generations:**
 - **802.11b, g, n, ac, ax** → Successive WLAN generations.
 - **802.11n, ac, ax** rebranded as **WiFi 4, 5, 6**.
 - **802.11af, ah** → Longer range, targeted for **IoT, sensors, metering**.
- **Common features:**
 - Same **frame format** across versions (backward compatible).
 - Medium access via **CSMA/CA**; 802.11ax also supports **centralized scheduling**.
- **Differences at physical layer:**
 - Operate in **2.4 GHz band** (shared with phones, microwaves; more interference) and **5 GHz band** (shorter range, more multipath issues).
 - **MIMO antennas** in 802.11n/ac/ax: multiple antennas for sending/receiving, improving throughput.
 - **Beamforming (ac/ax)**: base stations can target transmissions to reduce interference and extend range.
 - **Multi-user transmission**: 802.11ac/ax can serve multiple stations simultaneously.
- **Performance:** Published data rates assume **ideal conditions** (close to base station, no interference). Real-world rates are typically lower → **“Your Mileage May Vary (YMMV)”**.

Wireless LAN Architecture

Wireless LAN Architecture



Summary of 802.11 Wireless LAN Architecture (Figure 7.7)

- **Basic Service Set (BSS):**
 - Core building block of 802.11 architecture.

- Consists of one or more wireless stations plus a central **Access Point (AP)**.
- **Access Point (AP):**
 - Functions as the central base station for a BSS.
 - In larger networks, multiple APs connect to a switch or router, which then connects to the Internet.
 - In home setups, usually one AP + router (often combined in a single device).
- **MAC Addresses:**
 - Each wireless station has a **6-byte MAC address**, stored in its network interface card (NIC).
 - Each AP also has its own MAC address.
 - These addresses are assigned by IEEE and are globally unique (at least in theory).
- **Infrastructure Wireless LANs:**
 - WLANs that use APs are called **infrastructure mode LANs**.
 - The APs and supporting network devices form the “infrastructure” that connects wireless stations to the wider Internet.

Summary of 802.11 Association, SSID, and Channels

- **Association Requirement:**

- Before sending/receiving data, each wireless station must **associate** with an Access Point (AP).
- This process is used across all 802.11 standards (b, g, n, ac, ax).

- **SSID (Service Set Identifier):**

- Each AP is assigned a **one- or two-word SSID** by the administrator.
- SSIDs identify APs (e.g., Wi-Fi names shown on a phone).

- **Channel Numbers (2.4 GHz band):**

- 802.11 operates in **2.4–2.4835 GHz**, an **85 MHz band**.
- Defines **11 partially overlapping channels**.
- Two channels are **non-overlapping only if ≥ 4 apart**.
- Channels **1, 6, and 11** form the only set of three non-overlapping channels.

- **Practical Deployment:**

- To maximize throughput, an administrator can place **three APs in the same location** on channels 1, 6, and 11.
- Each AP connects to a switch → enabling an **aggregate max transmission rate $\approx 3\times$** that of a single AP.

WiFi Jungle, Scanning, Association, and Authentication

- **WiFi Jungle Concept**

- A **WiFi jungle** occurs when a device detects strong signals from multiple APs at once.
 - Example: a café where you can connect to the café's AP, but also pick up signals from nearby apartments.
 - Each AP typically belongs to a **different IP subnet** and is configured on its own **channel**.
 - To access the Internet, the device must **associate with exactly one AP**—this creates a “virtual wire” between device and AP.
 - Only the **associated AP** forwards/receives frames to/from the device.
-

- **Beacon Frames & Passive Scanning**

- Each AP periodically transmits **beacon frames**, containing its:
 - **SSID** (network name shown in WiFi lists).
 - **MAC address**.
 - A wireless device scans across the 11 possible channels in the 2.4 GHz band (or channels in the 5 GHz band), listening for beacon frames.
 - Multiple APs may be found on the same or different channels.
 - After collecting beacon info, the device (or user) chooses one AP.
 - Selection is usually based on **strongest signal strength**, but this may not always yield the best performance (strong AP could be overloaded).
-

- **Active Scanning**

- Instead of waiting for beacon frames, the device sends a **probe request**.
 - All APs in range reply with a **probe response**.
 - The device then chooses one AP.
 - **Association handshake:**
 1. Device sends an **association request frame** to chosen AP.
 2. AP replies with an **association response frame**.
 - This handshake is necessary because multiple APs may respond to the probe, but the device can only finalize association with one AP.
 - Analogy: similar to how a **DHCP client** may receive offers from multiple DHCP servers and then chooses one.
-

- **Subnet Integration**

- After associating with an AP, the device must obtain an IP address within that AP's subnet.
 - This is usually done by sending a **DHCP discovery message** through the AP.
 - DHCP server replies with an IP lease.
 - Once assigned, the device is seen by the outside Internet simply as another host in that subnet.
-

- **Authentication Methods**

- Many networks require **authentication** before association is finalized.

- Approaches include:
 - **MAC address filtering** → only devices with approved MAC addresses can connect.
 - **Username and password login** → common in cafés and hotspots.
- APs often rely on a **centralized authentication server** for these decisions.
 - Protocols: **RADIUS (RFC 2865)** or **DIAMETER (RFC 6733)**.
 - AP forwards credentials to the server, which decides whether to permit access.
 - Advantage: lowers AP complexity and allows one server to serve multiple APs.
- The **802.11i protocol** (covered in the next chapter) standardizes advanced security (encryption + authentication).

👉 **In short:** In a WiFi jungle, your device must **discover APs** (via passive or active scanning), **select and associate with one AP**, **obtain an IP address in that subnet**, and often **authenticate** (MAC-based or login-based). Only then can it exchange data with the Internet through that AP.

Multiple Access and Link-Layer Acknowledgments

- **Need for Multiple Access Protocol:**
 - Once a wireless device associates with an AP, multiple **stations** (devices + AP) may want to transmit on the same channel.

- A protocol is required to coordinate access.
 - 802.11 designers chose a **random access protocol** → **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance), inspired by Ethernet's CSMA/CD.
-

- **CSMA in Both Ethernet and WiFi:**

- **Carrier sensing:** station checks if the channel is busy before transmitting.
 - **Ethernet (CSMA/CD):** uses **collision detection** (abort if collision sensed).
 - **WiFi (CSMA/CA):** uses **collision avoidance** (no collision detection).
-

- **Why WiFi Doesn't Use Collision Detection:**

1. **Hardware limitation:** Detecting collisions requires sending and receiving simultaneously. Since transmitted signals are much stronger than received ones, building hardware to do both is costly.
 2. **Hidden terminal and fading:** Even if hardware allowed collision detection, many collisions would remain undetectable.
-

- **Consequence:**

- Once a WiFi station begins transmitting, it must **send the entire frame** (no abort mid-transmission).
- Collisions with long frames waste significant bandwidth → hence collision avoidance is necessary.

- **802.11 Link-Layer Acknowledgments (ARQ):**

- Wireless links have higher error rates than wired → acknowledgment mechanism is required.
- Process:
 1. Sender transmits a frame.
 2. Receiver checks CRC.
 3. If frame is correct, receiver waits a **Short Inter-frame Spacing (SIFS)**, then sends an **ACK frame** back.
 4. If sender receives ACK → success.
 5. If no ACK arrives within timeout → sender assumes error/collision and **retransmits**, using CSMA/CA rules.
 6. After a fixed number of failed retransmissions → frame is discarded.

👉 **Key Idea:** Unlike Ethernet, WiFi cannot detect collisions directly, so it **avoids collisions** with CSMA/CA and ensures reliability with **link-layer acknowledgments + retransmissions**.

Would you like me to also expand next on the **collision avoidance techniques (RTS/CTS, random backoff, etc.)** that 802.11 uses after this acknowledgment step?

CSMA/CA Protocol

1. Idle Channel Case:

- If a station has a frame to send and senses the channel **idle**, it waits a short period called **DIFS (Distributed Inter-frame Space)**.
- Then it **transmits the entire frame**.

2. Busy Channel Case:

- If the channel is **busy**, the station selects a **random backoff value** (using **binary exponential backoff**).
- Countdown only happens while the channel is idle, and pauses if the channel becomes busy again.

3. Transmission Start:

- When the backoff counter reaches **0**, the station transmits its **entire frame**.
- Then it waits for an **ACK**.

4. Acknowledgment:

- If an **ACK is received** → transmission was successful. If more frames remain, the station restarts at step 2.
- If **no ACK** is received → station assumes collision/error, re-enters **backoff phase**, this time with a **larger contention window**.

Why CSMA/CA Uses Backoff Even When Channel is Idle

● Ethernet's CSMA/CD:

- Stations transmit immediately once the channel is free.

- If collisions occur, they are quickly detected, and transmissions are aborted mid-frame (wasting little bandwidth).
 - **WiFi's CSMA/CA:**
 - Collisions **cannot be detected** mid-frame → once a frame collides, the **entire frame is wasted**.
 - To reduce this risk, WiFi stations **don't transmit immediately** after the channel is idle.
 - Instead, they **randomly back off** to avoid simultaneous starts.
-

Example Scenario (3 Stations)

- Two stations (A and B) wait because a third station (C) is transmitting.
 - When C finishes:
 - In Ethernet → A and B both transmit immediately → collision occurs → both abort.
 - In WiFi → A and B both **enter random backoff**.
 - If they choose different values, one transmits first; the other freezes its countdown and waits → **collision avoided**.
 - If values are close, or A and B are **hidden from each other**, a collision can still happen.
-

👉 Key Idea:

802.11's CSMA/CA **prioritizes avoiding collisions** (since they are very costly in wireless), while Ethernet's CSMA/CD **relies on detecting and aborting collisions quickly**.

RTS/CTS for Dealing with Hidden Terminals

- **Hidden terminal problem:**
 - Two stations (H1 and H2) are both associated with the same AP but cannot hear each other due to fading.
 - If H1 is transmitting and H2 starts transmitting without realizing it, a **collision** occurs at the AP.
 - This wastes the entire time of both transmissions, since 802.11 cannot stop a frame mid-transmission.
- **RTS/CTS solution (optional in 802.11):**
 - Before sending a data frame, a station may transmit a **Request to Send (RTS)** frame to the AP.
 - The RTS includes the **duration of the upcoming DATA + ACK exchange**.
 - The AP, after receiving RTS, responds with a **Clear to Send (CTS)** frame.
 - The CTS serves two purposes:
 1. **Grants permission** to the sender (H1) to transmit its DATA frame.

2. **Informs other stations** within the AP's range (including hidden ones like H2) to remain silent for the reserved duration.

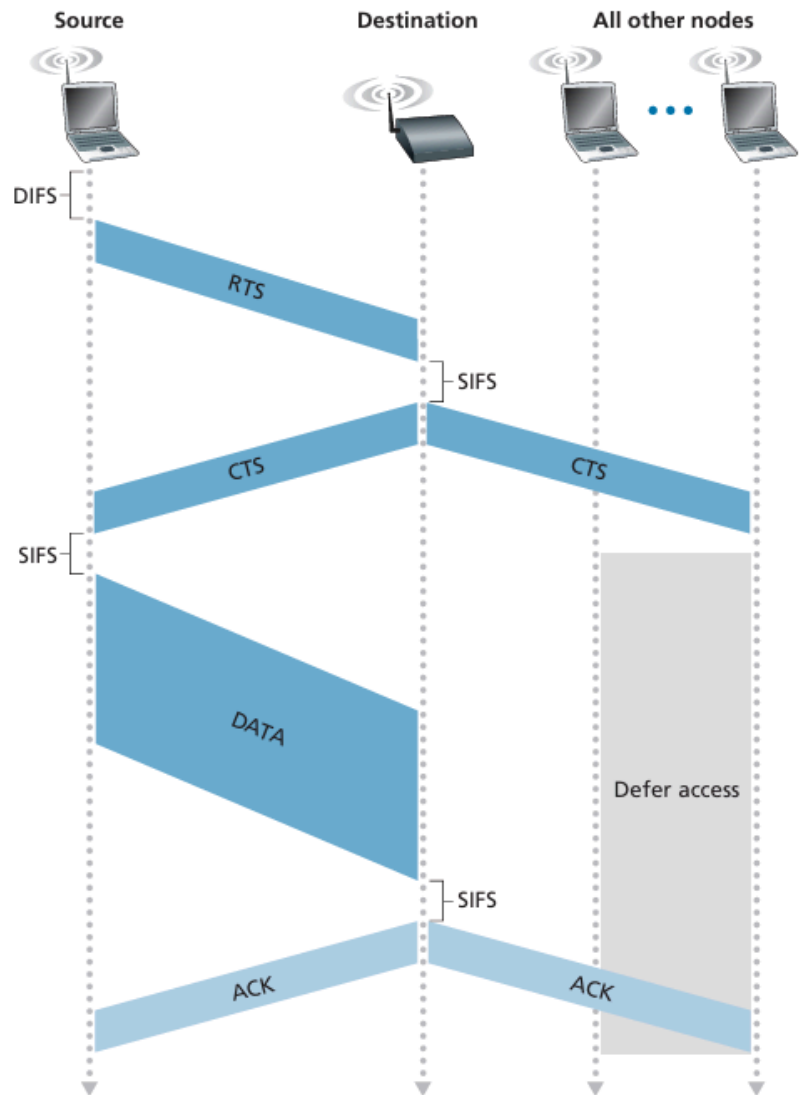


Figure 7.12 ♦ Collision avoidance using the RTS and CTS frames

- **Result:**

- H1 sends RTS → AP replies with CTS → H2 overhears CTS and stays quiet.
- H1 then transmits DATA, AP acknowledges with ACK.
- Collisions from hidden terminals are avoided, channel time is saved, and efficiency is improved.

👉 In short, **RTS/CTS adds a lightweight handshake** that reserves the channel, preventing hidden stations from interfering during data transmission.

PROBLEMS

P6. In step 4 of the CSMA/CA protocol, a station that successfully transmits a frame begins the CSMA/CA protocol for a second frame at step 2, rather than at step 1. What rationale might the designers of CSMA/CA have had in mind by having such a station not transmit the second frame immediately (if the channel is sensed idle)?

Why CSMA/CA Restarts at Step 2 (Backoff) Instead of Step 1

- **Fairness Concern:**
 - If a station that just transmitted were allowed to send another frame immediately whenever the channel is idle, it could **dominate the channel**.
 - Other stations waiting with backoff counters would be repeatedly “starved” since they never get a chance to access the channel.
- **Equal Opportunity for All Stations:**
 - By forcing the successful station to **enter backoff again (Step 2)**, even if the channel is idle, all stations get a fair chance.

- The previously successful station competes with others on equal footing in the next contention cycle.
 - **Collision Avoidance:**
 - Random backoff after each transmission reduces the likelihood that the same station transmits continuously, which could lead to more frequent collisions for other stations.
-

👉 **Rationale:** The rule prevents a single station from monopolizing the medium and ensures **fair access sharing** among all wireless stations.

Scenario Example: Why CSMA/CA Restarts at Step 2

Case 1 – If station could restart at Step 1 (no backoff):

- Suppose **Station A**, **Station B**, and **Station C** all have frames to send.
 - A wins the first contention (random backoff smallest), transmits its frame, and gets an ACK.
 - Now the channel is idle. If A could **immediately send another frame** (Step 1 rule), it transmits again.
 - B and C keep waiting → A keeps sending frame after frame.
 - Result: **Station A monopolizes the channel**, starving B and C.
-

Case 2 – With Step 2 rule (mandatory backoff):

- After A's successful transmission, it **re-enters random backoff** (Step 2).

- Meanwhile, B and C also have backoff counters running.
 - Example:
 - A chooses backoff = 10 slots.
 - B chooses backoff = 2 slots.
 - C chooses backoff = 7 slots.
 - Channel becomes idle → B's counter reaches 0 first, so **B transmits next**.
 - A and C freeze their counters while B is transmitting.
 - After B finishes, C (backoff = 7) gets its turn before A (backoff = 10).
 - All stations eventually transmit → **fairness maintained**.
-

Conclusion

By restarting at **Step 2 (random backoff)**, CSMA/CA:

- Prevents one station from **hogging the channel**.
 - Ensures **fair chance** for all stations.
 - Still minimizes collisions, since each station waits a randomized time before sending again.
-

- P5. Suppose there are two ISPs providing WiFi access in a particular café, with each ISP operating its own AP and having its own IP address block.
- Further suppose that by accident, each ISP has configured its AP to operate over channel 11. Will the 802.11 protocol completely break down in this situation? Discuss what happens when two stations, each associated with a different ISP, attempt to transmit at the same time.
 - Now suppose that one AP operates over channel 1 and the other over channel 11. How do your answers change?

a. Both APs on the Same Channel (Channel 11)

- **What happens:**
 - Both APs and their associated stations share the **same physical channel (same frequency band)**.
 - From 802.11's perspective, this looks like a **single wireless LAN** with multiple APs and stations.
 - When a station from ISP 1 and a station from ISP 2 transmit at the same time, their frames **collide** just like two stations in one WLAN.
 - **CSMA/CA** helps reduce collisions: stations sense the channel, back off, and retry.
 - Link-layer ACKs ensure lost frames are retransmitted.
 - **Does the protocol break down?**
 - **No.** 802.11 still works, but **throughput is reduced** because the two ISPs (which are meant to be separate) are now contending for the same channel bandwidth.
 - Both ISPs' customers experience **slower performance** due to more contention and retransmissions.
-

b. APs on Different Channels (Channel 1 and Channel 11)

- **What happens:**

- The two APs now operate on **non-overlapping frequency bands** (1 and 11).
- Transmissions on one channel do **not interfere** with transmissions on the other.
- From the stations' perspective, it's as if there are **two independent wireless LANs** coexisting in the same physical space.

- **Impact:**

- A station associated with ISP 1 (channel 1) can transmit simultaneously with a station associated with ISP 2 (channel 11) without collision.
- **No unnecessary contention** → each network gets its full channel capacity.
- **Performance is much better** than in case (a).

✓ **In summary:**

- **(a) Same channel (11):** Protocol still works, but both ISPs' networks interfere and contend → lower throughput.
- **(b) Different channels (1 & 11):** Truly independent operation → no interference, each ISP's WLAN performs normally.